The MOND acceleration scale as the cosmic dynamical acceleration: a scaling–MOND cosmology and its bridges to the dark sector

Carl Zimmerman
Independent Researcher

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Abstract

We examine the proposal that the Milgromian acceleration scale is the cosmic dynamical acceleration, $a_0 = \frac{c}{2}\sqrt{G\rho_c} = cH(z)/Z$ with $Z = 2\sqrt{8\pi/3} \simeq 5.789$. The evolving-scale hypothesis itself is Milgrom's (2014); our contribution is an explicit covariant realization of it and a proof of its CMB-safety. Algebraically $a_0 = cH/Z$ re-expresses the long-known coincidence $a_0 \approx cH_0$ through the Friedmann relation; its coefficient-free consequence is that the scale *evolves*, $a_0(z) = a_0(0)\,E(z)$. Fitting $a_0 \propto E(z)^p$ to the recent compilation (SPARC; Vărăşteanu 2025; MUSE–DARK III 2026) gives $p = 0.80 \pm 0.17$; constant a_0 is excluded at 5σ in the nominal fit, but a jackknife and an inter-method systematic (the 1.7σ discrepancy between the two near-coincident local anchors) reduce this to $\simeq 2\sigma$, so we treat the trend as a suggestive hint, not a detection. Working from the Aether–Scalar–Tensor (AeST) action we promote the scale to the aether expansion, $a_0 = c\theta/3Z$, and show by order-counting—reinforced by the identity $\delta q^{00} = 0$ that survives perturbing the aether itself—that a_0 is absent from the *linear* cosmological perturbations; the running therefore leaves the linear CMB and $P(k)$ exactly invariant, which we confirm by direct integration of the Einstein–Boltzmann system ($r_s = 144.3$ Mpc, $\ell_A = 301.7$). We are explicit about the boundaries: the $O(1)$ coefficient Z is a posit; the Standard-Model constants do not follow; and the apparent-a_0 evolution is itself reproduced by ΛCDM hydrodynamic simulations, so the amplitude alone does not discriminate. The one prediction that *does*—a redshift-weakening external field effect, $\eta = g_{\rm ext}/a_0(z) \propto 1/E(z)$, distinct from both ΛCDM and constant-a_0 MOND—is forecast to require a dedicated $z \gtrsim 4$ campaign.

I. INTRODUCTION

The Milgromian scale $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, below which galaxy dynamics depart systematically from the Newtonian expectation, is numerically close to the cosmic acceleration cH_0 [1, 4]. This proximity, $a_0 \approx cH_0/(2\pi)$, has been noted for forty years and is usually treated as either a coincidence or a clue that a_0 is set by cosmology [2]. Here we take the second view in a specific form, write the scale as a cosmic dynamical acceleration, derive its one falsifiable consequence, confront that with data, and map—honestly—how far the picture can be pushed toward a relativistic theory of the dark sector. We are equally explicit (Sec. VII) about what the construction does *not* do.

II. THE PREMISE AND ITS COEFFICIENT

We write the acceleration scale as half the surface gravity of the cosmic free-fall density,

$$a_0 = \frac{c}{2}\sqrt{G\rho_c} = \frac{c^2}{2R}, \qquad R = \sqrt{\tfrac{8\pi}{3}}\,\frac{c}{H} = c\,t_{\rm ff}, \tag{1}$$

where $t_{\rm ff} = (G\rho_c)^{-1/2}$ is the dynamical time of the critical-density medium. Using the Friedmann relation $H^2 = \frac{8\pi}{3}G\rho_c$, Eq. (1) is equivalent to

$$a_0 = \frac{cH}{Z}, \qquad Z = 2\sqrt{\tfrac{8\pi}{3}} = \sqrt{\tfrac{32\pi}{3}} \simeq 5.789. \tag{2}$$

The factor $\sqrt{8\pi/3}$ is the Friedmann free-fall-to-Hubble ratio and is not free; the remaining factor of two—the Schwarzschild reading $c^2/2R$—is the only undetermined element.

A physical reading sharpens *which* scale enters. $R = c\,t_{\rm ff}$ is the gravitational *free-fall* horizon—the radius at which the dynamical time $t_{\rm ff}$ of the critical medium equals the light-crossing time—as opposed to the *expansion* horizon c/H; the two differ by exactly the Friedmann factor, $R = \sqrt{8\pi/3}\,(c/H) \simeq 2.9\,c/H$. The same surface-gravity reading $c^2/2R$ applied at the *expansion* horizon gives $a_0 = cH/2$ ($Z = 2$, $a_0 \simeq 3.3 \times 10^{-10}$), which the data exclude at $\simeq 2.7\sigma$; that a_0 is fixed instead by the *collapse* rate $\sqrt{G\rho}$—as befits a gravitational acceleration set by the matter that sources it, not by the expansion rate—recovers Milgrom's observation that a_0 marks where a system's dynamical time reaches the cosmic time. This selects the density *form* of Eq. (1) and reduces the whole coefficient to a single statement, "a_0 is the surface gravity of the cosmic free-fall horizon"; it does not by itself fix the factor of two, to which we return in Sec. VI.

The density/surface-gravity *form* of Eq. (1) appears to be non-standard (the literature uses cH_0 or $\sqrt{\Lambda}$, not $\sqrt{G\rho}$), but it is a re-expression of a known coincidence rather than new dynamics, and the coefficient is not unique: the de Sitter-horizon readings of Milgrom ($Z = 2\pi$) and Verlinde ($Z \simeq 6$) [8] fit the same data equally well. We return to the status of Z in Sec. VI.

III. THE FALSIFIABLE PREDICTION AND THE DATA

Because a_0 tracks a cosmic density, it evolves—a possibility Milgrom himself raised [3]; and because Z cancels in the ratio, the prediction is coefficient-free and independent of the

Hubble tension:

$$\frac{a_0(z)}{a_0(0)} = E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{3}$$

Testing such evolution through the Tully–Fisher zero point was proposed early as a modified-gravity probe by Gnedin [6]. We fit $a_0(z) = A\,E(z)^p$ to the compilation in Table I, marginalizing the normalization A. The exponent is $p = 0.80 \pm 0.17$ (1σ); the model comparison gives $\chi^2 = 27$ for constant a_0, 3.8 for $E(z)$, and 27.5 for the matter-only law $(1+z)^{3/2}$. Thus in the nominal fit constant a_0 and the matter-only branch are each excluded at $\simeq 5\sigma$, while $p = 1$ is consistent at 1.1σ. This significance is fragile, and we state so plainly. A jackknife shows the rejection of constant a_0 rests almost entirely on the single $z \simeq 0.9$ point—dropping it leaves only 1.2σ—and the 1.7σ discrepancy between the two near-coincident local anchors (1.20 versus 1.69 at $z \simeq 0$, which cannot reflect genuine evolution over $\Delta z = 0.05$) evidences an inter-method systematic $\sigma_{\rm sys} \simeq 0.28$ (same units); folding $\sigma_{\rm sys}$ into all points reduces the rejection of constant a_0 to $\simeq 2\sigma$. We therefore regard the trend as a suggestive hint, not a detection, dominated by the local-anchor systematic rather than the redshift law. A single clean deep-MOND measurement at $z = 3$ with 3% precision would sharpen the exponent to $p \simeq 0.99 \pm 0.04$.

We emphasize two honest caveats. First, an evolving radial-acceleration relation is *also* expected in ΛCDM: hydrodynamic simulations (Magneticum, Tian *et al.* [11]) find the apparent best-fit a_0 rising by $\simeq 3$ from $z = 0$ to $z = 2.3$—matching $E(2.3) = 3.46$—with no fundamental evolution. The data thus favor “evolving over constant,” not the present framework over ΛCDM. Second, the dispersion-based dynamical masses of de Graaff *et al.* [10] at $z \simeq 6$ reach $M_{\rm dyn}/M_\star \sim 40$; these are *not* accounted for by the evolving scale, whose deep-MOND boost $\sqrt{a_0(z)/g_{\rm bar}}$ reaches only ~ 3 for such compact systems (which lie near the Newtonian regime at their epoch, $g_{\rm bar} \gtrsim a_0(z)$). They are a puzzle for MOND, evolving or not, and we do not cite them as support.

IV. THE OVER-CONSTRAINED WEB

The content of Eq. (2) is a single dimensionless invariant,

$$\frac{a_0}{cH(z)} = \frac{1}{Z} = 0.173 \quad \text{(every epoch)}, \tag{4}$$

TABLE I. The $a_0(z)$ compilation used in the fit. a_0 in $10^{-10}\,\mathrm{m\,s^{-2}}$. Provenance is tabulated in the repository (Data Availability).

z	a_0	source
0.00	1.20 ± 0.26	SPARC—McGaugh, Lelli & Schombert 2016 [5]
0.05	1.69 ± 0.13	Vărăşteanu *et al.* 2025 [12]
0.90	2.38 ± 0.11	MUSE–DARK III 2026 [13]

projected by dimensional analysis into different units—an acceleration (the radial-acceleration relation), a length $\ell = c^2/a_0 = ZR_H$, a surface density $\Sigma = a_0/G$, a velocity (the baryonic Tully–Fisher zero point), a temperature T_{dS}/Z, and a time (the cosmic free-fall time). From the one premise about a dozen relations follow as forced consequences (e.g. $H_0 = Za_0/c$; the de Sitter floor $a_0(0)\sqrt{\Omega_\Lambda}$; the Tully–Fisher and Faber–Jackson zero-points $\propto 1/E(z)$). The structure is *over-constrained*: the same a_0 must simultaneously fit the SPARC scale, the Λ floor $a_0(0)\sqrt{\Omega_\Lambda}$, and the intermediate-redshift points (Vărăşteanu, MUSE–DARK)—several determinations across cosmic time keyed to one number (the relation $H_0 = Za_0/c$ is an identity, not an independent check). This motivates a sharp filter: a genuine relation off Eq. (2) is z-invariant (a_0/cH stays $1/Z$), whereas a coincidence between cosmic numbers is not. The same test that retains these edges *rejects*, for example, the present-epoch identity $2\sin^2\theta_W \simeq \Omega_m/\Omega_\Lambda$, whose ratio drifts as $(1+z)^{-3}$—a why-now coincidence, not a law.

V. RELATIVISTIC COMPLETION AND CMB-SAFETY

A covariant home is the Aether–Scalar–Tensor (AeST) theory of Skordis & Zlosnik [7], the one relativistic MOND theory that reproduces the CMB and matter power spectra. Its action is

$$S = \int d^4x\, \frac{\sqrt{-g}}{16\pi\tilde{G}} \Big[R - \tfrac{K_B}{2} F^{\mu\nu}F_{\mu\nu} + 2(2-K_B)J^\mu\nabla_\mu\phi \\ - (2-K_B)\mathcal{Y} - \mathcal{F}(\mathcal{Y},\mathcal{Q}) - \lambda(A^\mu A_\mu + 1)\Big] + S_m[g], \quad (5)$$

with a unit-timelike vector ($A^\mu A_\mu = -1$), the spatial and temporal kinetic scalars $\mathcal{Y} = q^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi$ and $\mathcal{Q} = A^\mu\nabla_\mu\phi$ ($q_{\mu\nu} = g_{\mu\nu} + A_\mu A_\nu$), and a free function $\mathcal{F}$. The MOND scale enters the *spatial* sector, $\mathcal{F} \to [2\lambda_s/(1+\lambda_s)a_0]\,\mathcal{Y}^{3/2}$ as $\nabla\phi \to 0$, whereas the CMB-fitting, dark-matter-mimicking mode is carried by the *temporal* sector, $\mathcal{K}(\mathcal{Q}) = -2\Lambda + \mathcal{K}_2(\mathcal{Q} - \mathcal{Q}_0)^2$, which gives $\bar{\rho} \propto a^{-3}$ and contains no a_0.

We promote the constant a_0 to $a_0(z) = c\,\theta/(3Z)$, where $\theta = \nabla_\mu A^\mu$ is the aether expansion ($= 3H$ on a Friedmann–Robertson–Walker background); this reproduces Eq. (2) and is a minimal, covariant modification because θ is already present in Eq. (5). The key result is an order-counting statement. On the homogeneous background $\bar{\phi} = \bar{\phi}(t)$ is purely temporal, and because $q^{\mu\nu}$ projects orthogonal to A^μ one has $\bar{\mathcal{Y}} = 0$ and $\delta\mathcal{Y} = 0$ at linear order. The one term the standard order-counting omits—$\delta q^{\mu\nu}\nabla_\mu\bar{\phi}\,\nabla_\nu\bar{\phi}$, surviving only as $\delta q^{00}(\dot{\bar{\phi}})^2$—also vanishes identically, because the unit-timelike constraint forces $\delta A^0 = -\Psi$ and hence $\delta q^{00} = \delta g^{00} + 2A^0\delta A^0 = 2\Psi - 2\Psi = 0$. Thus robustly

$$\mathcal{Y} = O(\delta\phi^2), \qquad \mathcal{Y}^{3/2} = O(\delta\phi^3). \tag{6}$$

The a_0-bearing term is therefore third order in perturbations: it makes no contribution to the second-order action and hence none to the *linear* equations of motion, which are governed by the analytic term $-(2 - K_B)\mathcal{Y}$ and by $\mathcal{K}(\mathcal{Q})$, both a_0-independent. Consequently

$$a_0 \to a_0(z) \;\Rightarrow\; \{C_\ell,\, P(k)\}_{\rm linear} \text{ exactly invariant.} \tag{7}$$

We verified this with a direct integration of the linear Einstein–Boltzmann system: the running-a_0 effect on the linear spectra is numerically zero, and the background acoustic physics is recovered correctly ($r_s = 144.3$ Mpc, $\ell_A = 301.7$, against the Planck values 144.4 Mpc and $\simeq 301$). The running of a_0 thus enters only at nonlinear/quasi-static order. A Planck-precision C_ℓ amplitude calculation with the full vector sector is left to a dedicated Boltzmann code; it would reproduce the known AeST fit and does not alter the linear invariance just established. The $\mathcal{Y}^{3/2}$ non-analyticity makes the nonlinear matching delicate; the leading order-counting is robust.

VI. DARK-SECTOR UNIFICATION, AND THE COEFFICIENT

In Eq. (5) the single temporal function $\mathcal{K}(\mathcal{Q})$ carries *both* dark sectors: the dust mode ($\mathcal{K}_2(\mathcal{Q} - \mathcal{Q}_0)^2 \to \bar{\rho} \propto a^{-3}$, mimicking cold dark matter) and the cosmological constant (-2Λ). The evolving scale ties them: $a_0(z)$ floors at $a_0(0)\sqrt{\Omega_\Lambda}$, the Λ value. Dark matter and dark energy are, in this reading, one evolving scale at its two ends rather than two substances. The claim is falsifiable through a_0-cosmography: $H(z) = Z\,a_0(z)/c$ reconstructs the expansion from galaxy dynamics. The present data give $H(0) \simeq 71.5$ (within the H_0 band) and a $\sim 25\%$ excess at $z \simeq 0.9$—the same anchor-driven tension seen in Sec. III. A clean a_0 at $z > 2$ converts this into a measurement of the dark-energy equation of state from the dark-matter sector.

The coefficient. Horizon thermodynamics [8, 9] robustly yields the *evolution* $a_0 \propto H$ and the order of magnitude, but does not pin the $O(1)$ coefficient: distinct, individually defensible matchings give $Z = 1$ (Unruh=de Sitter temperature), $Z = 2\pi$ (Milgrom), $Z \simeq 6$ (Verlinde), and $Z = 5.789$ (the present density form). The collapse-horizon reading of Sec. II fixes the *form* and the scale $\sqrt{G\rho}$, but not the factor of two: the mass enclosed within R is $8\pi/3$ times the Schwarzschild mass for R, so R is a causal free-fall horizon, *not* a Schwarzschild horizon, and the surface-gravity factor $c^2/2R$ is borrowed from the Schwarzschild form. The factor of two therefore remains a posit, on the same footing as the 2π and 6 of the entropy readings. The physically motivated values cluster within 8.5%, inside the $\sim 20\%$ systematic on a_0, so the data cannot select among them. This costs the framework nothing, because the falsifiable prediction, Eq. (3), is Z-independent.

VII. SCOPE: WHAT IS NOT CLAIMED

The construction is a candidate description of the *dark sector* (gravity, dark matter, dark energy), not a theory of everything, and we state the boundary plainly because related notes have not. The Standard-Model constants do *not* follow from Eq. (2). A search over $\sim 3.4 \times 10^4$ simple formulae built from Z matches an arbitrary $O(100)$ target to 0.004% roughly 20% of the time; 52 of 64 such "derivations" contain no Z at all; and a randomly chosen number reproduces α^{-1} and the lepton/baryon mass ratios to the same precision as $32\pi/3$. Retrodictions of this kind carry ≈ 0 bits of evidence and form no part of the present framework. Likewise we do not claim a first-principles value for Z (Sec. VI) nor a Planck-precision CMB fit (Sec. V). Stating these limits is what keeps the surviving result—Eqs. (1)–(3) and the systematics-limited $\simeq 2\sigma$ preference for evolving over constant a_0—trustworthy.

VIII. THE DECISIVE TEST AND CONCLUSION

Pinning the exponent p by measuring a_0 at $z > 2$ tests evolution against constant a_0, but—because ΛCDM's *apparent* a_0 also rises as $E(z)$ (Sec. III)—it does *not* by itself separate the present framework from ΛCDM. The deep-MOND consequence cascade ($M_{\rm dyn}/M_\star \propto \sqrt{E}$, $\sigma \propto E^{1/4}$, the Tully–Fisher zero point $\propto -\log E$) shares this degeneracy: every channel follows from the radial-acceleration knee at $a_0(z)$, and ΛCDM, with apparent $a_0 \simeq a_0(0)E(z)$, returns the same powers.

The one observable that discriminates is the *external field effect*. With $a_0 \to a_0(z)$, a galaxy in a fixed environment $g_{\rm ext}$ has $\eta = g_{\rm ext}/a_0(z) \propto 1/E(z)$: the EFE *weakens* with redshift, so high-z galaxies in dense environments behave more like isolated deep-MOND systems. This is distinct from ΛCDM (no host effect on the internal relation) *and* from constant-a_0 MOND (constant EFE). Solving the MOND EFE numerically, the distinctive signal—the change across redshift in the embedded-versus-isolated mass-discrepancy offset—is $\simeq 0.1$ dex, below the per-galaxy scatter ($\simeq 0.25$–0.4 dex from kinematics, stellar and gas masses, and intrinsic scatter). A 3σ detection requires ~ 600–1600 *extended* (deep-MOND), environment-classified, kinematically-resolved galaxies at $z \gtrsim 4$—a dedicated multi-cycle JWST/ALMA(/ELT) program, not a single pointing.

In summary: $a_0 = \frac{c}{2}\sqrt{G\rho_c} = cH(z)/Z$ re-expresses Milgrom's coincidence in a density form; its evolving consequence $a_0 \propto E(z)$—an idea due to Milgrom [3]—is realized here covariantly in AeST, is provably CMB-safe at linear order, and is favored over constant a_0 by current data at $\simeq 2\sigma$. The coefficient remains a posit, the Standard-Model sector is untouched, and the amplitude is ΛCDM-degenerate. The framework's one distinctive, falsifiable signature is the redshift-weakening EFE, whose measurement we have forecast.

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Data and code availability. All calculations, the $a_0(z)$ compilation with its per-point provenance, and the linear Einstein–Boltzmann solver are reproducible from the public repository `https://github.com/carlzimmerman/zimmerman-formula` (directory `real_research/`; scripts `a0_decisive_pipeline.py`, `REAL_WEB.py`, `bridge1_linear_boltzmann.py`, `bridge1_aest_equat` `bridge2_coefficient_thermodynamics.py`, `bridge3_dark_sector_unification.py`, `false_discovery` `can_another_number_do_it.py`).

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